

Question

Heating powdered metallic thorium in a nitrogen atmosphere to above 500°C produces a reddish-brown powder. This reddish-brown powder is a thorium nitride, which exhibits good diamagnetism and electrical insulation. In the nitride crystal, both Th and N atoms have two chemical environments. The Th atoms are in a close-packed arrangement, with a stacking sequence of $\dots ABABCBCACABABCBCAC \dots$; all tetrahedral voids or all octahedral voids between adjacent close-packed layers are occupied by N atoms; the ratio of these two types of N atoms is $1 : 1$. The unit cell parameters are $a = b = 387 \text{ pm}$; $c = 2739 \text{ pm}$; $\alpha = \beta = 90^{\circ}$; $\gamma = 120^{\circ}$. Which of the following options is correct?

- A.** $\frac{1}{2}$ of the octahedral voids are occupied, $\frac{1}{2}$ of the tetrahedral voids are occupied.
- B.** The density of the crystal is 7.030 g/cm^3
- C.** Crystals have Th atoms with coordination number 6 only.
- D.** There are two types of Th atoms with different coordination numbers in the crystal, which are 6 and 7, respectively. The number of atoms with the two different coordination numbers is the same.
- E.** The chemical composition of the smallest unit of the crystal that repeats with translational periodicity is 1 formula unit.
- F.** The chemical composition of the smallest unit of this crystal that repeats with translational periodicity is 3 chemical formulas.
- G.** There are exactly 3 correct answers among options A to F.
- H.** Only 2 of options A to F are correct.

Answer

Correct Answer: **E**

Detailed Explanation

Because the ratio of the number of octahedral voids to tetrahedral voids in two adjacent close-packed layers is $1 : 2$, the ratio of the number of nitrogen atoms filling the two types of voids is $1 : 1$,

There are 9 layers of Th atoms in the unit cell, corresponding to 9 close-packed layers. Each layer can be filled with one octahedral void or two tetrahedral voids. Assume that x layers are filled with octahedral voids and $(9 - x)$ layers are filled with tetrahedral voids. Therefore, $1 \times x = 2 \times (9 - x)$, solving for $x = 6$, there are 6 octahedral void layers (6 octahedral voids filled), and 3 tetrahedral void layers (6 tetrahedral voids filled).

Therefore, $2/3$ of the octahedral void layers and $1/3$ of the tetrahedral void layers are filled with nitrogen atoms. There are 9 Th atoms in the unit cell, forming 9 octahedral voids and 18 tetrahedral voids; N fills 6 octahedral voids and 6 tetrahedral voids, that is, there are 12 N atoms in the unit cell, and the chemical formula of the crystal is Th_3N_4 .

CHECKPOINT

1 PTS

The ratio of the number of nitrogen atoms filling the two types of voids is $1 : 1$, so $2/3$ of the octahedral void layers and $1/3$ of the tetrahedral void layers are filled with nitrogen atoms

Option **A** is incorrect.

There are 9 Th atoms in a unit cell, corresponding to 9 close-packed layers, of which 3 layers are N -filled tetrahedral void layers, and 6 layers are N -filled octahedral void layers. The void layers are evenly distributed. Then 3 Th will be located between two layers of N -filled octahedral void layers, with a coordination number of 6, and the other 6 Th will be located between a layer of N -filled octahedral void layers and a layer of N -filled tetrahedral void layers, with a coordination number of 7. Therefore, there are 3 Th atoms with a coordination number of 6 and 6 Th atoms with a coordination number of 7.

CHECKPOINT

1 PTS

There are 3 *Th* atoms with a coordination number of 6 and 6 *Th* atoms with a coordination number of 7.

Options **C**, **D** are incorrect.

The smallest unit of the crystal that is periodically repeated by translation is the lattice point, which contains a structural motif. There are 9 layers of *Th* atoms in a unit cell, ... *ABA/BCB/CAC*... Divided into groups of every 3 layers of *Th* atoms, the smallest translation repeating unit can be obtained, which is 1 *Th*₃*N*₄, which is 1 chemical formula.

CHECKPOINT

1 PTS

The smallest translation repeating unit, which is 1 *Th*₃*N*₄, which is 1 chemical formula.

Option **E** is correct, option **F** is incorrect.

$$\rho = \frac{ZM}{N_A V} = \frac{3 \times (232.0 \times 3 + 14.01 \times 4)}{6.022 \times 10^{23} \times 387^2 \times 2739 \times \sin 120^\circ \times 10^{-30}} = 10.55 \text{ (g/cm}^3\text{)}$$

CHECKPOINT

1 PTS

$$\rho = \frac{ZM}{N_A V} = 10.55 \text{ (g/cm}^3\text{)}$$

Option **B** is incorrect

Options **G**, **H** are incorrect